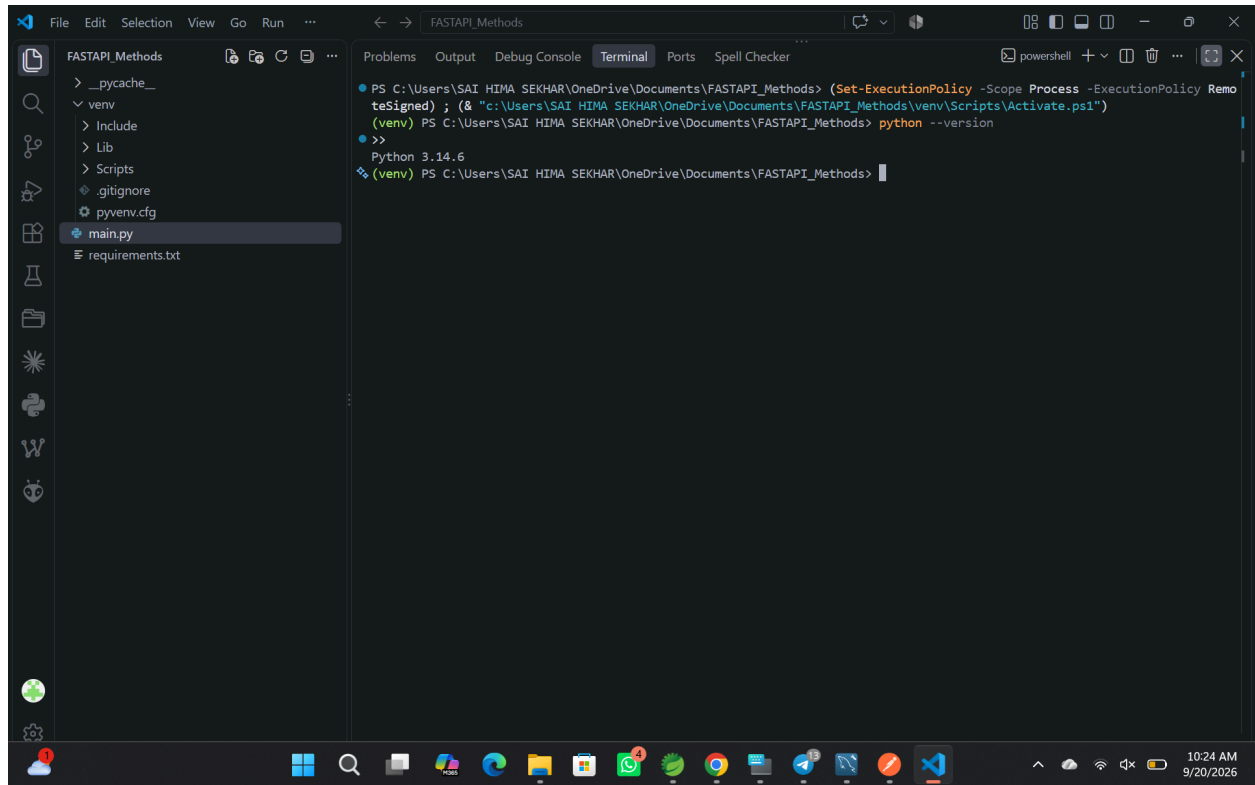


FASTAPI

Check Python



The screenshot shows a Visual Studio Code (VS Code) interface with a terminal window open. The terminal is running PowerShell commands to activate a virtual environment and check the Python version.

```
PS C:\Users\SAI HIMA SEKHAR\OneDrive\Documents\FASTAPI_Methods> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\SAI HIMA SEKHAR\OneDrive\Documents\FASTAPI_Methods\venv\Scripts\Activate.ps1")
(venv) PS C:\Users\SAI HIMA SEKHAR\OneDrive\Documents\FASTAPI_Methods> python --version
Python 3.14.6
(venv) PS C:\Users\SAI HIMA SEKHAR\OneDrive\Documents\FASTAPI_Methods>
```

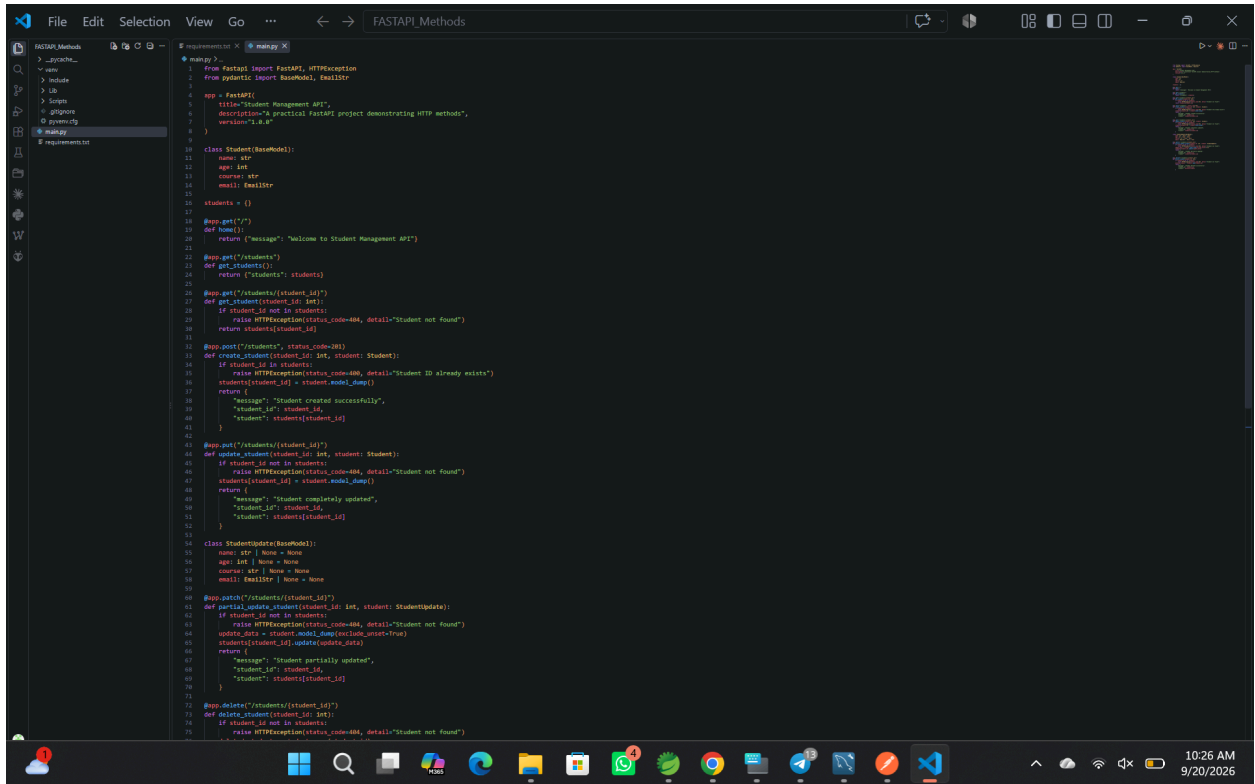
The left sidebar shows the file explorer with the following structure:

- FASTAPI_Methods
 - > __pycache__
 - > venv
 - > Include
 - > Lib
 - > Scripts
 - .gitignore
 - pyvenv.cfg
 - main.py
 - requirements.txt

The bottom status bar shows the time as 10:24 AM on 9/20/2026.

FASTAPI

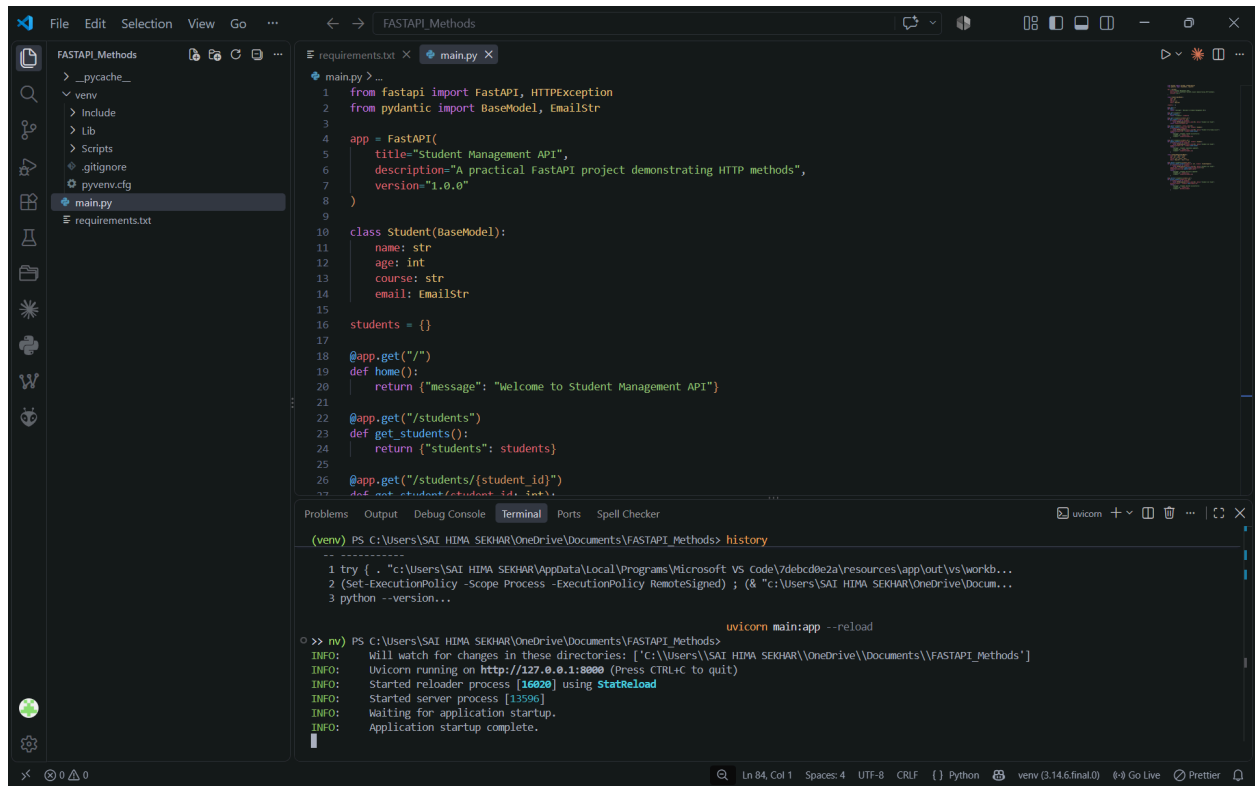
Create [main.py](#)



```
1 from fastapi import FastAPI, HTTPException
2 from pydantic import BaseModel, EmailStr
3
4 app = FastAPI()
5 title = "Student Management API"
6 description = "A practical FastAPI project demonstrating HTTP methods"
7 version = "1.0.0"
8
9
10 class Student(BaseModel):
11     name: str
12     age: int
13     course: str
14     email: EmailStr
15
16 students = []
17
18 @app.get("/")
19 def home():
20     return {"message": "Welcome to Student Management API!"}
21
22 @app.get("/students")
23 def get_students():
24     return {"students": students}
25
26 @app.get("/students/{student_id}")
27 def get_student(student_id: int):
28     if student_id not in students:
29         raise HTTPException(status_code=404, detail="Student not found")
30     return students[student_id]
31
32 @app.post("/students", status_code=201)
33 def create_student(student_id: int, student: Student):
34     if student_id in students:
35         raise HTTPException(status_code=400, detail="Student ID already exists")
36     students[student_id] = student.model_dump()
37     return {
38         "message": "Student created successfully",
39         "student_id": student_id,
40         "student": students[student_id]
41     }
42
43 @app.put("/students/{student_id}")
44 def update_student(student_id: int, student: Student):
45     if student_id not in students:
46         raise HTTPException(status_code=404, detail="Student not found")
47     students[student_id] = student.model_dump()
48     return {
49         "message": "Student completely updated",
50         "student_id": student_id,
51         "student": students[student_id]
52     }
53
54 class StudentUpdate(BaseModel):
55     name: str | None = None
56     age: int | None = None
57     course: str | None = None
58     email: EmailStr | None = None
59
60 @app.patch("/students/{student_id}")
61 def partial_update_student(student_id: int, student: StudentUpdate):
62     if student_id not in students:
63         raise HTTPException(status_code=404, detail="Student not found")
64     update_data = student.model_dump(exclude_unset=True)
65     students[student_id].update(update_data)
66     return {
67         "message": "Student partially updated",
68         "student_id": student_id,
69         "student": students[student_id]
70     }
71
72 @app.delete("/students/{student_id}")
73 def delete_student(student_id: int):
74     if student_id not in students:
75         raise HTTPException(status_code=404, detail="Student not found")
```

FASTAPI

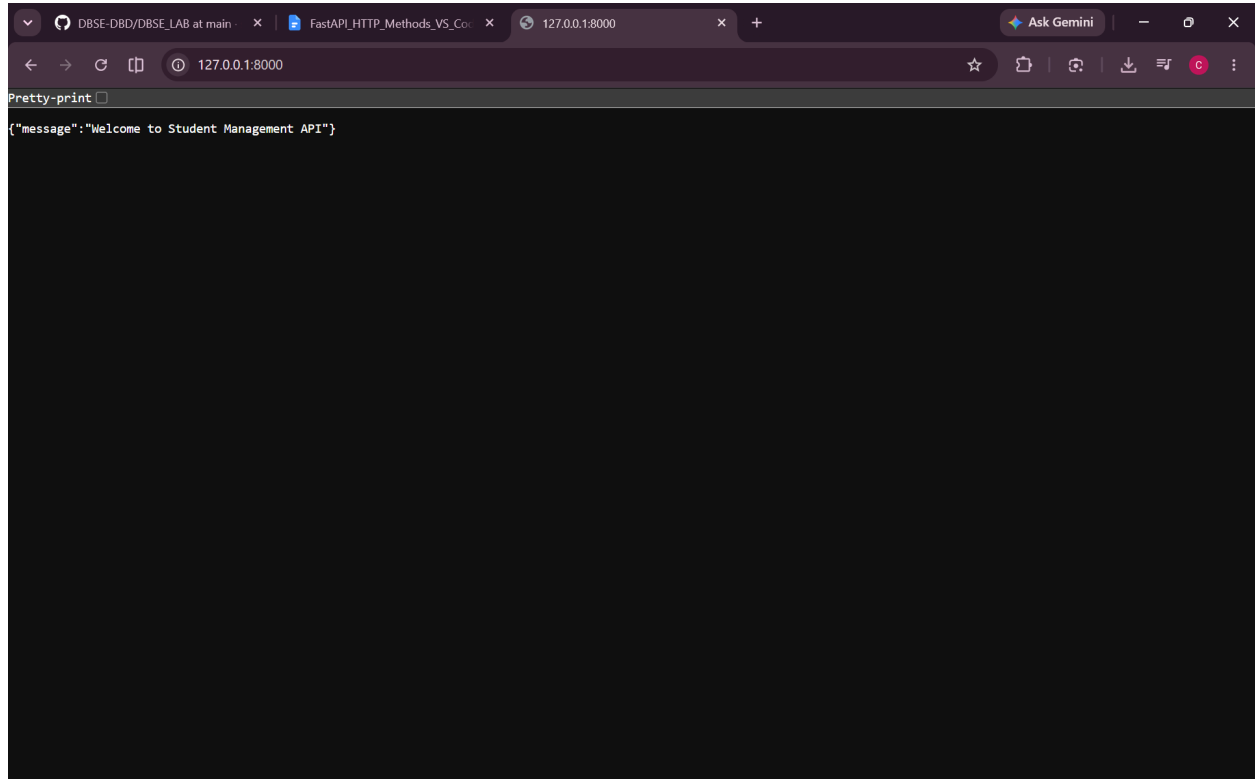
Start the FastAPI Server



The screenshot shows the VS Code editor with the `main.py` file open. The file contains the following code:

```
1 from fastapi import FastAPI, HTTPException
2 from pydantic import BaseModel, EmailStr
3
4 app = FastAPI(
5     title="Student Management API",
6     description="A practical FastAPI project demonstrating HTTP methods",
7     version="1.0.0"
8 )
9
10 class Student(BaseModel):
11     name: str
12     age: int
13     course: str
14     email: EmailStr
15
16 students = {}
17
18 @app.get("/")
19 def home():
20     return {"message": "Welcome to Student Management API"}
21
22 @app.get("/students")
23 def get_students():
24     return {"students": students}
25
26 @app.get("/students/{student_id}")
27 def get_student(student_id: int):
28     return {"student": students[student_id]}
```

The terminal window at the bottom shows the command `uvicorn main:app --reload` being executed. The output indicates that the application is running on `http://127.0.0.1:8000` and is using `StatReload` for reloading.



The screenshot shows a web browser window with the address `127.0.0.1:8000`. The browser displays the response from the FastAPI API, which is a JSON object: `{"message": "Welcome to Student Management API"}`. The browser's developer tools are open, showing the response in the console.

FASTAPI

Open Swagger UI

The screenshot shows the Swagger UI for the 'Student Management API' (version 1.0.0, OAS 3.1). The API is described as 'A practical FastAPI project demonstrating HTTP methods'. The 'default' section lists the following endpoints:

- GET** / Home
- GET** /students Get Students
- POST** /students Create Student
- GET** /students/{student_id} Get Student
- PUT** /students/{student_id} Update Student
- PATCH** /students/{student_id} Partial Update Student
- DELETE** /students/{student_id} Delete Student

The browser's taskbar at the bottom shows the time as 10:30 AM on 9/20/2026.

GET

The screenshot shows the details for the **GET** / Home endpoint. The 'Parameters' section indicates 'No parameters'. The 'Responses' section shows a successful response (200) with the following details:

- Code:** 200
- Response body:** {"message": "Welcome to Student Management API"}
- Response headers:** content-length: 47, content-type: application/json, date: Sun, 20 Sep 2026 10:30:18 GMT, server: uvicorn

The 'Responses' table at the bottom shows a 'Successful Response' (200) with a media type of 'application/json' and an example value of 'string'.

FASTAPI

The screenshot shows a web browser with the URL `127.0.0.1:8000/docs#/default/get_students_students_get`. The interface is for the `GET /students` endpoint. The **Parameters** section shows "No parameters". The **Responses** section shows a successful response with status code 200. The response body is a JSON object: `{ "students": [] }`. The response headers are: `content-length: 15`, `content-type: application/json`, `date: Sun, 20 Sep 2026 05:01:46 GMT`, and `server: uvicorn`. The **Responses** table shows a "Successful Response" with status code 200 and media type `application/json`.

POST

The screenshot shows a web browser with the URL `127.0.0.1:8000/docs#/default/create_student_students_post`. The interface is for the `POST /students` endpoint. The **Parameters** section shows a required parameter `student_id` with the value `1`. The **Request body** section shows a required JSON object: `{ "name": "string", "age": 0, "course": "string", "email": "user@example.com" }`. The **Responses** section shows a successful response with status code 200. The response body is a JSON object: `{ "name": "string", "age": 0, "course": "string", "email": "user@example.com" }`. The **Responses** table shows a "Successful Response" with status code 200 and media type `application/json`.

FASTAPI

Request URL: `http://127.0.0.1:8000/students/{student_id}-1`

Server response

Code: 201

Response body

```
{
  "message": "Student created successfully",
  "student_id": 1,
  "student": {
    "name": "string",
    "age": 0,
    "course": "string",
    "email": "user@example.com"
  }
}
```

Response headers

```
content-length: 138
content-type: application/json
date: Sun, 20 Sep 2026 10:33:08 GMT
server: uvicorn
```

Responses

Code	Description	Links
201	Successful Response	No links
422	Validation Error	No links

Media type:

Controls: Accept header

Example Value / Schema

```
"string"
```

PUT

PUT /students/{student_id} Update Student

Parameters

Name	Description
student_id ^{required}	1

Request body ^{required}

Media type:

Edit Value / Schema

```
{
  "name": "string",
  "age": 0,
  "course": "string",
  "email": "user@example.com"
}
```

Execute Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/students/1' \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "name": "string",
    "age": 0,
    "course": "string",
    "email": "user@example.com"
  }'
```

FASTAPI

DELETE

The screenshot displays a web browser window with the address bar showing the URL `127.0.0.1:8000/docs#/default/delete_student_students_student_id_delete`. The browser tabs include "DBSE-DB8D/DBSE_LAB at main", "FastAPI_HTTP_Methods_VS_Co...", and "Student Management API - Sw...".

The main content area shows a REST client interface for a **DELETE** endpoint: `/students/{student_id}` with the description "Delete Student".

Parameters:

Name	Description
student_id * required	
Integer	
(path)	

The parameter `student_id` is set to `1`. Below the parameters, there are "Execute" and "Clear" buttons.

Responses:

Curl:

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/students/1' \
  -H 'accept: */*'
```

Request URL:

```
http://127.0.0.1:8000/students/1
```

Server response:

Code: 200

Response body:

```
{
  "message": "Student deleted successfully",
  "student_id": 1,
  "student": {
    "name": "string",
    "age": 0,
    "course": "string",
    "email": "user@example.com"
  }
}
```

Response headers:

```
content-length: 138
content-type: application/json
date: Sun, 20 Sep 2026 10:36:27 GMT
server: uvicorn
```

The bottom of the screen shows a Windows taskbar with various application icons and a system clock indicating 10:36 AM on 9/20/2026.